

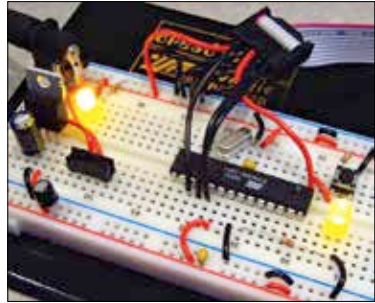
tronic circuits. A good guide to use a breadboard is available at <https://learn.sparkfun.com/tutorials/how-to-use-a-breadboard>

- **Single strand wires**

- These are wires with one thick strand and can be used to make connections on a breadboard with ease.

- **Relays**

- We use mechanical switches at our home to switch lights ON or OFF.
- The relay helps perform this action electrically.



A breadboard for electronic prototyping

- **Bluetooth / Wi-Fi module**

- When building a home automation system, communication between various modules in a home is important, such communication is possible via use of Bluetooth and Wi-Fi modules.
- Bluetooth and Wi-Fi modules also help to connect an appropriately programmed smartphone to connect to the home automation system installed in that home. This way you can ensure the system behaves as necessary by monitor it or overriding it as necessary.

- **Sensors**

- Sensors act as an interface between the real world and the digital world. They help the microcontroller or micro-processor to understand various real world parameters.



A relay used as electronic switch

Software components

Software components for a home automation system can be divided into two parts:

- **Microcontroller /**

microprocessor software

- Microcontroller / microprocessor contains logic to decide if a device should be turned ON or OFF.
- This software connects various devices like air conditioning to sen-